

AI & Machine Learning for Civil Engineers

From engineering data to prediction, explainability and uncertainty, in eight hands-on sessions.

08 Live sessions**90** min Per session**04** weeks Course duration **Runs 100% on Google Colab**

Nothing to install, works even on a Chromebook

 Built for beginners

No prior Python or machine learning required

 Live code-along on Civil Engineering datasets

Materials · Buildings & Energy · Transport · Environment · Geotechnics

 Hands-on notebooks, cross-domain assignments & solutions

Plus certificate & recordings (till 2 weeks after the course)

INSTRUCTOR**Dr. Abdul Waris**

Research Associate

Universität Siegen (University of Siegen)

Former Prime Minister's Research Fellow

Ph.D. in Geotechnical Engineering, IIT Hyderabad

 Research Excellence Award 2026, IIT Hyderabad Invited talks at Oregon State University (USA) and a Faculty Development Programme at D.Y. Patil (Pune, India) Resource person, five-day ML workshop at NIT Srinagar

Course outline

01 Python for Civil Engineers

AI/ML basics, Colab, NumPy & pandas with engineering data

02 From Raw Data to ML-Ready

Explore, clean, visualize & split engineering data

03 Your First Prediction Model

Linear regression, error metrics & cross-validation

04 Trees & Ensemble Models

Decision Trees, Random Forest & boosting for nonlinear problems

05 Classification for Engineering Decisions

Logistic regression, confusion matrix, precision & recall

06 Artificial Neural Networks

Architecture, activations, training & overfitting

07 Explainable AI & Engineering Judgement

Feature importance, SHAP, and engineering interpretation

08 From Prediction to Probability

Monte Carlo, ML surrogates & probability of failure

 **Designed for Civil Engineers across disciplines** — Materials, Structures, Geotechnics, Transportation, Water & Environment.

 **Early Bird offer till 12 October**

Regular fees apply after 12 Oct

WHO SHOULD JOIN

B.Tech & M.Tech students / PhD scholars / Faculty / Practising engineers / Complete beginners to AI & ML

STARTS · LIMITED SEATS

**Saturday,
17 October 2026**

Scan to register · qudelastudios.com →

